

PARABANK

AUTOMATION TESTING FRAMEWORK

Complete Source Code Documentation

JAVA | SELENIUM | TESTNG | MAVEN | JENKINS | DOCKER

Prepared By	Pratha Khare
Project Type	BFSI Web Automation Capstone
Document	Eclipse-Style Complete Source Code
Prepared On	June 2026

Prepared from the supplied Eclipse Maven project archive

1. Introduction

The ParaBank Automation Testing Framework is a Java-based web automation project designed to validate core banking workflows in the ParaBank demonstration application. The solution uses Selenium WebDriver for browser interaction, TestNG for test management and assertions, Maven for build execution, and the Page Object Model for separation of page behaviour from test logic.

This source code document has been prepared directly from the supplied Eclipse project. The code is presented in an editor-style layout without artificial line numbering so that every class and configuration file appears similar to code copied from Eclipse.

The framework covers registration, login, account creation, account overview, fund transfer and bill payment. It also includes Excel-based test data handling, reusable wait and screenshot utilities, Extent reporting, Log4j2 logging, a TestNG listener, a Jenkins pipeline and a Docker execution definition.

2. Project Objectives

- Automate important functional workflows of a BFSI web application.
- Apply the Page Object Model to create maintainable and reusable automation code.
- Manage input data and generated credentials through Microsoft Excel using Apache POI.
- Generate execution evidence through screenshots, logs and Extent HTML reports.
- Execute the complete suite through TestNG, Maven, Jenkins and Docker.
- Maintain the project in an Eclipse-compatible Maven structure.

3. Project Profile

Project Name	ParaBank Automation Testing Framework
Prepared By	Pratha Khare
Application Under Test	ParaBank Demo Banking Application
Programming Language	Java
Automation Tool	Selenium WebDriver
Test Framework	TestNG
Build and Dependency Tool	Apache Maven
Framework Design	Page Object Model
Test Data	Apache POI with ParaBankData1.xlsx
Reporting	Extent Reports
Logging	Log4j2
CI/CD	Jenkins Pipeline
Containerization	Docker

4. Automated Modules

Module	Automation Coverage
Registration	Creates a new ParaBank customer with a dynamically generated username and saves credentials to Excel.
Login	Reads stored credentials from Excel and validates successful access to the customer home page.
Open Account	Creates a new savings account and validates the account-opened confirmation.
Account Overview	Navigates to the account summary and verifies that the Accounts Overview page is available.
Transfer Funds	Selects source and destination accounts, enters the transfer amount and submits the transaction.
Bill Payment	Uses Excel data to populate payee details and validates successful bill payment.

5. Framework Architecture

Base Layer — Initializes WebDriver, applies Docker-compatible Chrome options, opens the configured URL and closes the browser.

Page Layer — Contains locators and reusable actions for registration, login, account, transfer and bill payment pages.

Test Layer — Contains TestNG test classes, assertions and module execution logic.

Utility Layer — Provides configuration reading, Excel operations, random data, waits, screenshots and report management.

Integration Layer — Contains TestNG XML, Maven configuration, Jenkins pipeline and Dockerfile.

6. Source Inclusion Note

This document contains 32 readable source and configuration files from the supplied project archive. Generated build folders, compiled classes, HTML execution reports, log files, screenshots and other runtime output have not been reproduced as source code.

The Excel workbook ParaBankData1.xlsx is part of the project test-data package. Because it is a binary spreadsheet rather than source code, it is referenced in the project structure but is not printed as text in this document.

7. Eclipse Project Structure

```
ParaBank/
├── .classpath
├── .project
├── .settings/
│   ├── org.eclipse.jdt.core.prefs
│   └── org.eclipse.m2e.core.prefs
├── Dockerfile
├── Jenkinsfile
├── pom.xml
├── testng.xml
├── testdata/
│   └── ParaBankData1.xlsx
└── src/test/
    ├── java/
    │   ├── base/
    │   ├── listeners/
    │   ├── pages/
    │   ├── testcases/
    │   └── utilities/
    └── resources/
        ├── config.properties
        └── log4j2.xml
```

8. Source Code Index

No.	File Path	Lines
1	.classpath	38
2	.project	23
3	.gitignore	1
4	.settings/org.eclipse.jdt.core.prefs	15
5	.settings/org.eclipse.m2e.core.prefs	4
6	pom.xml	92
7	testng.xml	45
8	Dockerfile	22
9	Jenkinsfile	48
10	src/test/resources/config.properties	2
11	src/test/resources/log4j2.xml	36
12	src/test/java/base/BaseTest.java	47
13	src/test/java/listeners/TestListener.java	82
14	src/test/java/pages/AccountOverviewPage.java	28
15	src/test/java/pages/BillPayPage.java	78

No.	File Path	Lines
16	src/test/java/pages/HomePage.java	51
17	src/test/java/pages/LoginPage.java	42
18	src/test/java/pages/OpenAccountPage.java	63
19	src/test/java/pages/RegistrationPage.java	50
20	src/test/java/pages/TransferFundsPage.java	69
21	src/test/java/testcases/AccountOverviewTest.java	52
22	src/test/java/testcases/BillPayTest.java	86
23	src/test/java/testcases/LoginTest.java	45
24	src/test/java/testcases/OpenAccountTest.java	44
25	src/test/java/testcases/RegistrationTest.java	57
26	src/test/java/testcases/TransferFundsTest.java	45
27	src/test/java/utilities/ConfigReader.java	31
28	src/test/java/utilities/ExcelUtils.java	84
29	src/test/java/utilities/ExtentManager.java	32
30	src/test/java/utilities/RandomDataGenerator.java	25
31	src/test/java/utilities/ScreenshotUtil.java	42
32	src/test/java/utilities/WaitUtils.java	21

COMPLETE SOURCE CODE

Eclipse-style presentation • Original indentation preserved • No artificial line numbers

.classpath

FILE 01 OF 32

.classpath

```
<?xml version="1.0" encoding="UTF-8"?>
<classpath>
  <classpathentry kind="src" output="target/classes" path="src/main/java">
    <attributes>
      <attribute name="optional" value="true"/>
      <attribute name="maven.pomderived" value="true"/>
    </attributes>
  </classpathentry>
  <classpathentry excluding="*" kind="src" output="target/classes" path="src/main/resources">
    <attributes>
      <attribute name="maven.pomderived" value="true"/>
    </attributes>
  </classpathentry>
  <classpathentry kind="src" output="target/test-classes" path="src/test/java">
    <attributes>
      <attribute name="optional" value="true"/>
      <attribute name="maven.pomderived" value="true"/>
      <attribute name="test" value="true"/>
    </attributes>
  </classpathentry>
  <classpathentry excluding="*" kind="src" output="target/test-classes" path="src/test/resources">
    <attributes>
      <attribute name="maven.pomderived" value="true"/>
      <attribute name="test" value="true"/>
    </attributes>
  </classpathentry>
  <classpathentry kind="con" path="org.eclipse.jdt.launching.JRE_CONTAINER/org.eclipse.jdt.internal.debug.ui.launcher.StandardVMType/JavaSE-11">
    <attributes>
      <attribute name="maven.pomderived" value="true"/>
    </attributes>
  </classpathentry>
  <classpathentry kind="con" path="org.eclipse.m2e.MAVEN2_CLASSPATH_CONTAINER">
    <attributes>
      <attribute name="maven.pomderived" value="true"/>
    </attributes>
  </classpathentry>
  <classpathentry kind="output" path="target/classes"/>
</classpath>
```

Source: [Eclipse Maven project archive](#) | 38 lines

.project

FILE 02 OF 32

.project

```
<?xml version="1.0" encoding="UTF-8"?>
<projectDescription>
  <name>ParaBank</name>
  <comment></comment>
  <projects>
  </projects>
  <buildSpec>
    <buildCommand>
      <name>org.eclipse.jdt.core.javabuilder</name>
      <arguments>
      </arguments>
    </buildCommand>
    <buildCommand>
      <name>org.eclipse.m2e.core.maven2Builder</name>
      <arguments>
      </arguments>
    </buildCommand>
  </buildSpec>
  <natures>
    <nature>org.eclipse.jdt.core.javanature</nature>
    <nature>org.eclipse.m2e.core.maven2Nature</nature>
  </natures>
</projectDescription>
```

Source: Eclipse Maven project archive | 23 lines

.gitignore

FILE 03 OF 32

`.gitignore``/target/`*Source: Eclipse Maven project archive | 1 lines*

org.eclipse.jdt.core.prefs

FILE 04 OF 32

`.settings/org.eclipse.jdt.core.prefs`

```
eclipse.preferences.version=1
org.eclipse.jdt.core.compiler.codegen.inlineJsrBytecode=enabled
org.eclipse.jdt.core.compiler.codegen.targetPlatform=11
org.eclipse.jdt.core.compiler.codegen.unusedLocal=preserve
org.eclipse.jdt.core.compiler.compliance=11
org.eclipse.jdt.core.compiler.debug.lineNumber=generate
org.eclipse.jdt.core.compiler.debug.localVariable=generate
org.eclipse.jdt.core.compiler.debug.sourceFile=generate
org.eclipse.jdt.core.compiler.problem.assertIdentifier=error
org.eclipse.jdt.core.compiler.problem.enablePreviewFeatures=disabled
org.eclipse.jdt.core.compiler.problem.enumIdentifier=error
org.eclipse.jdt.core.compiler.problem.forbiddenReference=warning
org.eclipse.jdt.core.compiler.problem.reportPreviewFeatures=ignore
org.eclipse.jdt.core.compiler.release=disabled
org.eclipse.jdt.core.compiler.source=11
```

Source: Eclipse Maven project archive | 15 lines

org.eclipse.m2e.core.prefs

FILE 05 OF 32

`.settings/org.eclipse.m2e.core.prefs`

```
activeProfiles=  
eclipse.preferences.version=1  
resolveWorkspaceProjects=true  
version=1
```

Source: Eclipse Maven project archive | 4 lines

pom.xml

FILE 06 OF 32

pom.xml

```

<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
    http://maven.apache.org/xsd/maven-4.0.0.xsd">

  <modelVersion>4.0.0</modelVersion>

  <groupId>com.parabank</groupId>
  <artifactId>ParaBankAutomation</artifactId>
  <version>1.0-SNAPSHOT</version>
  <packaging>jar</packaging>

  <properties>
    <maven.compiler.source>11</maven.compiler.source>
    <maven.compiler.target>11</maven.compiler.target>
    <selenium.version>4.18.1</selenium.version>
    <testng.version>7.9.0</testng.version>
    <poi.version>5.2.5</poi.version>
  </properties>

  <dependencies>

    <!-- Selenium WebDriver -->
    <dependency>
      <groupId>org.seleniumhq.selenium</groupId>
      <artifactId>selenium-java</artifactId>
      <version>${selenium.version}</version>
    </dependency>

    <!-- WebDriverManager (auto-manages chromedriver/geckodriver) -->
    <dependency>
      <groupId>io.github.bonigarcia</groupId>
      <artifactId>webdrivermanager</artifactId>
      <version>5.7.0</version>
    </dependency>

    <!-- TestNG -->
    <dependency>
      <groupId>org.testng</groupId>
      <artifactId>testng</artifactId>
      <version>${testng.version}</version>
    </dependency>

    <!-- Apache POI (Excel) -->
    <dependency>
      <groupId>org.apache.poi</groupId>
      <artifactId>poi-ooxml</artifactId>
      <version>${poi.version}</version>
    </dependency>

    <!-- Commons IO (screenshots) -->
    <dependency>
      <groupId>commons-io</groupId>
      <artifactId>commons-io</artifactId>
      <version>2.15.1</version>
    </dependency>
  </dependencies>

```

```
<groupId>com.aventstack</groupId>
<artifactId>extentreports</artifactId>
<version>5.1.1</version>
</dependency>
<dependency>
  <groupId>org.apache.logging.log4j</groupId>
  <artifactId>log4j-api</artifactId>
  <version>2.24.3</version>
</dependency>

<dependency>
  <groupId>org.apache.logging.log4j</groupId>
  <artifactId>log4j-core</artifactId>
  <version>2.24.3</version>
</dependency>

</dependencies>

<build>
  <plugins>
    <plugin>
      <groupId>org.apache.maven.plugins</groupId>
      <artifactId>maven-surefire-plugin</artifactId>
      <version>3.2.5</version>
      <configuration>
        <suiteXmlFiles>
          <suiteXmlFile>testng.xml</suiteXmlFile>
        </suiteXmlFiles>
      </configuration>
    </plugin>
  </plugins>
</build>

</project>
```

Source: Eclipse Maven project archive | 92 lines

testng.xml

FILE 07 OF 32

testng.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE suite SYSTEM "https://testng.org/testng-1.0.dtd">

<suite name="ParaBank Automation Suite" verbose="1" parallel="false">
  <listeners>
    <listener class-name="listeners.TestListener"/>
  </listeners>
  <parameter name="browser" value="chrome"/>

  <test name="Registration Tests">
    <classes>
      <class name="testcases.RegistrationTest"/>
    </classes>
  </test>
  <test name="Login Tests">
    <classes>
      <class name="testcases.LoginTest"/>
    </classes>
  </test>
  <test name="Open Account Tests">
    <classes>
      <class name="testcases.OpenAccountTest"/>
    </classes>
  </test>
  <test name="Account Overview Tests">
    <classes>
      <class name="testcases.AccountOverviewTest"/>
    </classes>
  </test>

  <test name="Transfer Funds Tests">
    <classes>
      <class name="testcases.TransferFundsTest"/>
    </classes>
  </test>

  <test name="Bill Pay Tests">
    <classes>
      <class name="testcases.BillPayTest"/>
    </classes>
  </test>
</suite>
```

Source: Eclipse Maven project archive | 45 lines

Dockerfile

FILE 08 OF 32

Dockerfile

```
FROM maven:3.9.6-eclipse-temurin-21

RUN apt-get update && apt-get install -y \
    wget \
    gnupg \
    unzip

RUN wget -q -O - https://dl.google.com/linux/linux_signing_key.pub | \
    gpg --dearmor -o /usr/share/keyrings/google.gpg

RUN echo "deb [arch=amd64 signed-by=/usr/share/keyrings/google.gpg] http://dl.google.com/linux/chrome/deb/ stable main" \
    > /etc/apt/sources.list.d/google.list

RUN apt-get update && apt-get install -y google-chrome-stable

WORKDIR /app

COPY . .

RUN mvn dependency:resolve

CMD ["mvn", "test", "-Ddocker=true"]
```

Source: Eclipse Maven project archive | 22 lines

Jenkinsfile

FILE 09 OF 32

Jenkinsfile

```
pipeline {
  agent any

  tools {
    maven 'Maven3'
    jdk 'jdk25'
  }

  stages {
    stage('Checkout') {
      steps {
        git branch: 'master',
            url: 'https://github.com/pratha25/Para-Bank-Automation-.git'
      }
    }

    stage('Build') {
      steps {
        dir('ParaBank') {
          bat 'mvn clean compile'
        }
      }
    }

    stage('Run Tests') {
      steps {
        dir('ParaBank') {
          bat 'mvn test'
        }
      }
    }
  }

  post {
    always {
      archiveArtifacts artifacts: '**/target/surefire-reports/*, **/test-output/**/*, **/screenshots/**/*',
          allowEmptyArchive: true
    }

    success {
      echo 'Build Successful!'
    }

    failure {
      echo 'Build Failed!'
    }
  }
}
```

Source: Eclipse Maven project archive | 48 lines

config.properties

FILE 10 OF 32

`src/test/resources/config.properties``url=https://parabank.parasoft.com/parabank/index.htm
browser=chrome`*Source: Eclipse Maven project archive | 2 lines*

log4j2.xml

FILE 11 OF 32

src/test/resources/log4j2.xml

```
<?xml version="1.0" encoding="UTF-8"?>

<Configuration status="WARN">

  <Appenders>

    <Console name="Console"
      target="SYSTEM_OUT">

      <PatternLayout
        pattern="%d{HH:mm:ss} %-5level %msg%n"/>

    </Console>

    <File name="File"
      fileName="reports/automation.log">

      <PatternLayout
        pattern="%d %-5level %msg%n"/>

    </File>

  </Appenders>

  <Loggers>

    <Root level="info">

      <AppenderRef ref="Console"/>
      <AppenderRef ref="File"/>

    </Root>

  </Loggers>

</Configuration>
```

Source: Eclipse Maven project archive | 36 lines

BaseTest.java

FILE 12 OF 32

src/test/java/base/BaseTest.java

```
package base;

import java.time.Duration;
import org.openqa.selenium.chrome.ChromeOptions;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeDriver;
import org.testng.annotations.AfterMethod;
import org.testng.annotations.BeforeMethod;

import utilities.ConfigReader;

public class BaseTest {
    public WebDriver getDriver() {
        return driver;
    }
    protected WebDriver driver;

    @BeforeMethod
    public void setup() {

        ChromeOptions options = new ChromeOptions();

        String docker =
            System.getProperty("docker");

        if("true".equalsIgnoreCase(docker)) {

            options.addArguments("--headless=new");
            options.addArguments("--no-sandbox");
            options.addArguments("--disable-dev-shm-usage");
            options.addArguments("--window-size=1920,1080");
        }

        driver = new ChromeDriver(options);

        driver.manage().timeouts()
            .implicitlyWait(Duration.ofSeconds(10));

        driver.get(ConfigReader.getProperty("url"));
    }
    @AfterMethod
    public void tearDown() {

        if(driver!=null)
            driver.quit();
    }
}
```

Source: Eclipse Maven project archive | 47 lines

TestListener.java

FILE 13 OF 32

src/test/java/listeners/TestListener.java

```

package listeners;
import base.BaseTest;
import utilities.ScreenshotUtil;
import org.testng.ITestContext;
import org.testng.ITestListener;
import org.testng.ITestResult;
import com.aventstack.extentreports.ExtentReports;
import com.aventstack.extentreports.ExtentTest;

import utilities.ExtentManager;
public class TestListener implements ITestListener {
    ExtentReports extent =
        ExtentManager.getReport();

    ExtentTest test;
    @Override
    public void onTestStart(ITestResult result) {

        test = extent.createTest(
            result.getName());

        System.out.println(
            "STARTED : "
            + result.getName());
    }

    @Override
    public void onTestSuccess(ITestResult result) {

        test.pass("Test Passed");

        System.out.println(
            "PASSED : "
            + result.getName());
    }

    @Override
    public void onTestFailure(ITestResult result) {

        test.fail(result.getThrowable());

        try {

            BaseTest base =
                (BaseTest) result.getInstance();

            String screenshotPath =
                ScreenshotUtil.captureScreenshot(
                    base.getDriver(),
                    result.getName());
            System.out.println(
                "Screenshot Path = "
                + screenshotPath);
            test.addScreenCaptureFromPath(
                screenshotPath);

        } catch (Exception e) {

```

```
        e.printStackTrace();
    }

    System.out.println(
        "FAILED : "
        + result.getName());
}

@Override
public void onStart(ITestContext context) {

    System.out.println("Execution Started");
}

@Override
public void onFinish(ITestContext context) {

    extent.flush();

    System.out.println(
        "Execution Finished");
}
}
```

Source: Eclipse Maven project archive | 82 lines

AccountOverviewPage.java

FILE 14 OF 32

src/test/java/pages/AccountOverviewPage.java

```
package pages;

import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;

public class AccountOverviewPage {

    WebDriver driver;

    public AccountOverviewPage(WebDriver driver) {
        this.driver = driver;
    }

    By accountOverview =
        By.linkText("Accounts Overview");

    public void openAccountOverview() {

        driver.findElement(accountOverview)
            .click();
    }

    public boolean verifyAccountOverview() {

        return driver.getPageSource()
            .contains("Accounts Overview");
    }
}
```

Source: Eclipse Maven project archive | 28 lines

BillPayPage.java

FILE 15 OF 32

src/test/java/pages/BillPayPage.java

```
package pages;

import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;

public class BillPayPage {

    WebDriver driver;

    public BillPayPage(WebDriver driver) {

        this.driver = driver;
    }

    By billPayLink = By.linkText("Bill Pay");

    By payeeName = By.name("payee.name");
    By address = By.name("payee.address.street");
    By city = By.name("payee.address.city");
    By state = By.name("payee.address.state");
    By zipCode = By.name("payee.address.zipCode");
    By phone = By.name("payee.phoneNumber");
    By account = By.name("payee.accountNumber");
    By verifyAccount = By.name("verifyAccount");
    By amount = By.name("amount");

    By sendPayment =
        By.xpath("//input[@value='Send Payment']");

    public void payBill(
        String payeeNameValue,
        String addressValue,
        String cityValue,
        String stateValue,
        String zipCodeValue,
        String phoneValue,
        String accountValue,
        String verifyAccountValue,
        String amountValue) {

        driver.findElement(billPayLink).click();

        driver.findElement(payeeName)
            .sendKeys(payeeNameValue);

        driver.findElement(address)
            .sendKeys(addressValue);

        driver.findElement(city)
            .sendKeys(cityValue);

        driver.findElement(state)
            .sendKeys(stateValue);

        driver.findElement(zipCode)
            .sendKeys(zipCodeValue);

        driver.findElement(phone)
```

```
        .sendKeys(phoneValue);

    driver.findElement(account)
        .sendKeys(accountValue);

    driver.findElement(verifyAccount)
        .sendKeys(verifyAccountValue);

    driver.findElement(amount)
        .sendKeys(amountValue);

    driver.findElement(sendPayment)
        .click();
}
public boolean verifyBillPayment() {

    return driver.getPageSource()
        .contains("Bill Payment Complete");
}
}
```

Source: Eclipse Maven project archive | 78 lines

HomePage.java

FILE 16 OF 32

src/test/java/pages/HomePage.java

```
package pages;

import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;

public class HomePage {

    WebDriver driver;

    public HomePage(WebDriver driver) {
        this.driver = driver;
    }

    By logoutLink = By.linkText("Log Out");
    By accountOverviewLink = By.linkText("Accounts Overview");
    By transferFundsLink = By.linkText("Transfer Funds");
    By billPayLink = By.linkText("Bill Pay");

    public boolean isHomePageDisplayed() {

        return driver.findElement(logoutLink)
            .isDisplayed();
    }

    public void clickAccountOverview() {

        driver.findElement(accountOverviewLink)
            .click();
    }

    public void clickTransferFunds() {

        driver.findElement(transferFundsLink)
            .click();
    }

    public void clickBillPay() {

        driver.findElement(billPayLink)
            .click();
    }

    public void logout() {

        driver.findElement(logoutLink)
            .click();
    }
}
```

Source: Eclipse Maven project archive | 51 lines

LoginPage.java

FILE 17 OF 32

src/test/java/pages/LoginPage.java

```
package pages;

import java.time.Duration;

import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.support.ui.ExpectedConditions;
import org.openqa.selenium.support.ui.WebDriverWait;

public class LoginPage {

    WebDriver driver;

    public LoginPage(WebDriver driver) {
        this.driver = driver;
    }

    By username = By.name("username");

    By password = By.name("password");

    By loginButton =
        By.xpath("//input[@value='Log In']");

    public void login(String user, String pass) {

        WebDriverWait wait =
            new WebDriverWait(driver,
                Duration.ofSeconds(10));

        wait.until(
            ExpectedConditions.visibilityOfElementLocated(username));

        driver.findElement(username).clear();
        driver.findElement(username).sendKeys(user);

        driver.findElement(password).clear();
        driver.findElement(password).sendKeys(pass);

        driver.findElement(loginButton).click();
    }
}
```

Source: Eclipse Maven project archive | 42 lines

OpenAccountPage.java

FILE 18 OF 32

src/test/java/pages/OpenAccountPage.java

```
package pages;

import org.openqa.selenium.By;
import org.openqa.selenium.JavascriptExecutor;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.support.ui.Select;

public class OpenAccountPage {

    WebDriver driver;

    public OpenAccountPage(WebDriver driver) {
        this.driver = driver;
    }

    By openNewAccountLink =
        By.linkText("Open New Account");

    By accountType =
        By.id("type");

    By fromAccount =
        By.id("fromAccountId");

    By openAccountButton =
        By.xpath("//input[@value='Open New Account']");

    public void openNewAccountPage() {

        driver.findElement(openNewAccountLink)
            .click();
    }

    public void createNewAccount() {

        Select typeDropdown =
            new Select(driver.findElement(accountType));

        typeDropdown.selectByVisibleText("SAVINGS");

        Select accountDropdown =
            new Select(driver.findElement(fromAccount));

        accountDropdown.selectByIndex(0);

        WebElement button =
            driver.findElement(openAccountButton);

        JavascriptExecutor js =
            (JavascriptExecutor) driver;

        js.executeScript(
            "arguments[0].click();",
            button);
    }

    public boolean verifyAccountCreated() {
```

```
        return driver.getPageSource()  
                .contains("Account Opened");  
    }  
}
```

Source: Eclipse Maven project archive | 63 lines

RegistrationPage.java

FILE 19 OF 32

src/test/java/pages/RegistrationPage.java

```
package pages;

import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;

public class RegistrationPage {

    WebDriver driver;

    public RegistrationPage(WebDriver driver) {

        this.driver = driver;
    }

    By firstName = By.id("customer.firstName");
    By lastName = By.id("customer.lastName");
    By address = By.id("customer.address.street");
    By city = By.id("customer.address.city");
    By state = By.id("customer.address.state");
    By zip = By.id("customer.address.zipCode");
    By phone = By.id("customer.phoneNumber");
    By ssn = By.id("customer.ssn");
    By username = By.id("customer.username");
    By password = By.id("customer.password");
    By confirm = By.id("repeatedPassword");
    By register = By.xpath("//input[@value='Register']");

    public void registerUser(String fname,
                             String lname,
                             String user,
                             String pass) {

        driver.findElement(firstName).sendKeys(fname);
        driver.findElement(lastName).sendKeys(lname);
        driver.findElement(address).sendKeys("India");
        driver.findElement(city).sendKeys("Mahoba");
        driver.findElement(state).sendKeys("UP");
        driver.findElement(zip).sendKeys("210421");
        driver.findElement(phone).sendKeys("7080852802");
        driver.findElement(ssn).sendKeys("123456");

        driver.findElement(username).sendKeys(user);

        driver.findElement(password).sendKeys(pass);

        driver.findElement(confirm).sendKeys(pass);

        driver.findElement(register).click();
    }
}
```

Source: Eclipse Maven project archive | 50 lines

TransferFundsPage.java

FILE 20 OF 32

src/test/java/pages/TransferFundsPage.java

```
package pages;

import java.time.Duration;

import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.support.ui.ExpectedConditions;
import org.openqa.selenium.support.ui.Select;
import org.openqa.selenium.support.ui.WebDriverWait;

public class TransferFundsPage {

    WebDriver driver;

    public TransferFundsPage(WebDriver driver) {
        this.driver = driver;
    }

    // Locators
    By transferFundsLink = By.linkText("Transfer Funds");

    By amount = By.id("amount");

    By fromAccount = By.id("fromAccountId");

    By toAccount = By.id("toAccountId");

    By transferButton = By.xpath("//input[@value='Transfer']");

    public void transferFunds(String transferAmount) {

        driver.findElement(transferFundsLink).click();

        WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(10));

        wait.until(ExpectedConditions.visibilityOfElementLocated(amount));

        driver.findElement(amount).clear();
        driver.findElement(amount).sendKeys(transferAmount);

        Select from = new Select(driver.findElement(fromAccount));

        Select to = new Select(driver.findElement(toAccount));

        System.out.println("From Accounts Count : " + from.getOptions().size());

        System.out.println("To Accounts Count : " + to.getOptions().size());

        from.selectByIndex(0);

        if (to.getOptions().size() > 1) {
            to.selectByIndex(1);
        } else {
            to.selectByIndex(0);
        }

        driver.findElement(transferButton).click();
    }
}
```

```
}  
  
public String getTransferMessage() {  
    WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(10));  
    WebElement message = wait.until(ExpectedConditions.visibilityOfElementLocated(By.tagName("h1")));  
    return message.getText();  
}  
}
```

Source: Eclipse Maven project archive | 69 lines

AccountOverviewTest.java

FILE 21 OF 32

src/test/java/testcases/AccountOverviewTest.java

```
package testcases;

import org.testng.Assert;
import org.testng.annotations.Test;
import org.apache.logging.log4j.LogManager;
import org.apache.logging.log4j.Logger;
import base.BaseTest;
import pages.AccountOverviewPage;
import pages.LoginPage;
import utilities.ExcelUtils;

public class AccountOverviewTest extends BaseTest {
    private static final Logger logger =
        LogManager.getLogger(AccountOverviewTest.class);

    @Test
    public void accountOverviewTest() {

        logger.info("Account Overview Test Started");

        LoginPage loginPage =
            new LoginPage(driver);

        loginPage.login(
            ExcelUtils.getData(
                "LoginData",
                1,
                0),

            ExcelUtils.getData(
                "LoginData",
                1,
                1)
        );

        logger.info("Login Successful");

        AccountOverviewPage accountPage =
            new AccountOverviewPage(driver);

        accountPage.openAccountOverview();

        logger.info("Account Overview Page Opened");

        Assert.assertTrue(
            accountPage.verifyAccountOverview());

        logger.info("Account Overview Test Passed");
    }
}
```

Source: Eclipse Maven project archive | 52 lines

BillPayTest.java

FILE 22 OF 32

src/test/java/testcases/BillPayTest.java

```
package testcases;

import org.testng.Assert;
import org.testng.annotations.Test;

import base.BaseTest;
import pages.BillPayPage;
import pages.LoginPage;
import utilities.ExcelUtils;

public class BillPayTest extends BaseTest {

    @Test
    public void billPayTest() throws InterruptedException {

        LoginPage login =
            new LoginPage(driver);
        login.login(
            ExcelUtils.getData(
                "LoginData",
                1,
                0),

            ExcelUtils.getData(
                "LoginData",
                1,
                1));

        BillPayPage billPay =
            new BillPayPage(driver);

        billPay.payBill(

            ExcelUtils.getData(
                "BillPayData",
                1,
                0),

            ExcelUtils.getData(
                "BillPayData",
                1,
                1),

            ExcelUtils.getData(
                "BillPayData",
                1,
                2),

            ExcelUtils.getData(
                "BillPayData",
                1,
                3),

            ExcelUtils.getData(
                "BillPayData",
                1,
                4),
```

```
        ExcelUtils.getData(  
            "BillPayData",  
            1,  
            5),  
  
        ExcelUtils.getData(  
            "BillPayData",  
            1,  
            6),  
  
        ExcelUtils.getData(  
            "BillPayData",  
            1,  
            7),  
  
        ExcelUtils.getData(  
            "BillPayData",  
            1,  
            8)  
    );  
    Thread.sleep(5000);  
    Assert.assertTrue(  
        billPay.verifyBillPayment(),  
        "Bill Payment Failed");  
    }  
}
```

Source: Eclipse Maven project archive | 86 lines

LoginTest.java

FILE 23 OF 32

src/test/java/testcases/LoginTest.java

```
package testcases;

import org.testng.Assert;
import org.testng.annotations.Test;

import base.BaseTest;
import pages.HomePage;
import pages.LoginPage;
import utilities.ExcelUtils;

public class LoginTest extends BaseTest {

    @Test
    public void loginTest() throws InterruptedException {

        String username =
            ExcelUtils.getData(
                "LoginData",
                1,
                0);

        String password =
            ExcelUtils.getData(
                "LoginData",
                1,
                1);

        System.out.println("Username = " + username);
        System.out.println("Password = " + password);

        LoginPage loginPage =
            new LoginPage(driver);

        loginPage.login(username, password);

        Thread.sleep(3000);

        HomePage homePage =
            new HomePage(driver);

        Assert.assertTrue(
            homePage.isHomePageDisplayed(),
            "Login Failed");
    }
}
```

Source: Eclipse Maven project archive | 45 lines

OpenAccountTest.java

FILE 24 OF 32

src/test/java/testcases/OpenAccountTest.java

```
package testcases;

import org.testng.Assert;
import org.testng.annotations.Test;

import base.BaseTest;
import pages.LoginPage;
import pages.OpenAccountPage;
import utilities.ExcelUtils;

public class OpenAccountTest extends BaseTest {

    @Test(priority = 4)

    public void createAccount() {

        LoginPage login =
            new LoginPage(driver);

        login.login(

            ExcelUtils.getData(
                "LoginData",
                1,
                0),

            ExcelUtils.getData(
                "LoginData",
                1,
                1)

        );

        OpenAccountPage account =
            new OpenAccountPage(driver);

        account.openNewAccountPage();

        account.createNewAccount();

        Assert.assertTrue(
            account.verifyAccountCreated(),
            "Account Creation Failed");
    }
}
```

Source: Eclipse Maven project archive | 44 lines

RegistrationTest.java

FILE 25 OF 32

src/test/java/testcases/RegistrationTest.java

```
package testcases;

import org.testng.annotations.Test;

import base.BaseTest;
import pages.RegistrationPage;
import utilities.ExcelUtils;
import utilities.RandomDataGenerator;

public class RegistrationTest extends BaseTest {

    @Test
    public void registerUser() {

        driver.get("https://parabank.parasoft.com/parabank/register.htm");

        String username = RandomDataGenerator.getUsername();
        String password =
            ExcelUtils.getData(
                "RegistrationData",
                1,
                8);

        RegistrationPage reg = new RegistrationPage(driver);
        reg.registerUser(

            ExcelUtils.getData(
                "RegistrationData",
                1,
                0),

            ExcelUtils.getData(
                "RegistrationData",
                1,
                1),

            username,

            password
        );

        ExcelUtils.writeData(
            "LoginData",
            1,
            0,
            username);

        ExcelUtils.writeData(
            "LoginData",
            1,
            1,
            password);

        System.out.println("Saved Username : " + username);
        System.out.println("Saved Password : " + password);
    }
}
```

Source: Eclipse Maven project archive | 57 lines

TransferFundsTest.java

FILE 26 OF 32

src/test/java/testcases/TransferFundsTest.java

```
package testcases;

import org.testng.Assert;
import org.testng.annotations.Test;

import base.BaseTest;
import pages.LoginPage;
import pages.TransferFundsPage;
import utilities.ExcelUtils;

public class TransferFundsTest extends BaseTest {
    @Test
    public void transferFundsTest() throws InterruptedException {

        LoginPage loginPage =
            new LoginPage(driver);

        loginPage.login(
            ExcelUtils.getData(
                "LoginData",
                1,
                0),

            ExcelUtils.getData(
                "LoginData",
                1,
                1));

        TransferFundsPage transferPage =
            new TransferFundsPage(driver);

        transferPage.transferFunds(
            ExcelUtils.getData(
                "TransferData",
                1,
                0)

        );

        Thread.sleep(3000);

        Assert.assertTrue(true);
    }
}
```

Source: Eclipse Maven project archive | 45 lines

ConfigReader.java

FILE 27 OF 32

src/test/java/utilities/ConfigReader.java

```
package utilities;

import java.io.FileInputStream;
import java.util.Properties;

public class ConfigReader {

    static Properties prop;

    static {

        try {

            FileInputStream fis =
                new FileInputStream("src/test/resources/config.properties");

            prop = new Properties();

            prop.load(fis);

        } catch (Exception e) {

            e.printStackTrace();

        }

    }

    public static String getProperty(String key) {

        return prop.getProperty(key);

    }

}
```

Source: Eclipse Maven project archive | 31 lines

ExcelUtils.java

FILE 28 OF 32

src/test/java/utilities/ExcelUtils.java

```
package utilities;

import java.io.FileInputStream;
import java.io.FileOutputStream;

import org.apache.poi.xssf.usermodel.XSSFSheet;
import org.apache.poi.xssf.usermodel.XSSFWorkbook;

public class ExcelUtils {

    public static void writeData(
        String sheetName,
        int row,
        int cell,
        String value) {

        String path = "testdata/ParaBankData1.xlsx";

        try {

            FileInputStream fis =
                new FileInputStream(path);

            XSSFWorkbook wb =
                new XSSFWorkbook(fis);

            XSSFSheet sheet =
                wb.getSheet(sheetName);

            sheet.getRow(row)
                .getCell(cell)
                .setCellValue(value);

            fis.close();

            FileOutputStream fos =
                new FileOutputStream(path);

            wb.write(fos);

            fos.close();

            wb.close();

        } catch (Exception e) {

            e.printStackTrace();

        }

    }

    public static String getData(
        String sheetName,
        int row,
        int cell) {

        String path = "testdata/ParaBankData1.xlsx";

        try {
```

```
FileInputStream fis =
    new FileInputStream(path);

XSSFWorkbook wb =
    new XSSFWorkbook(fis);

XSSFSheet sheet =
    wb.getSheet(sheetName);

String value =
    sheet.getRow(row)
        .getCell(cell)
        .toString();

wb.close();

return value;
} catch (Exception e) {
    e.printStackTrace();
    return "";
}
}
```

Source: Eclipse Maven project archive | 84 lines

ExtentManager.java

FILE 29 OF 32

src/test/java/utilities/ExtentManager.java

```
package utilities;

import com.aventstack.extentreports.ExtentReports;
import com.aventstack.extentreports.reporter.ExtentSparkReporter;

public class ExtentManager {

    private static ExtentReports extent;

    public static ExtentReports getReport() {

        if (extent == null) {

            ExtentSparkReporter spark =
                new ExtentSparkReporter(
                    "reports/ExtentReport.html");

            spark.config()
                .setReportName("ParaBank Automation Report");

            extent = new ExtentReports();

            extent.attachReporter(spark);

            extent.setSystemInfo(
                "Tester",
                "Pratha Khare");

        }

        return extent;

    }
}
```

Source: Eclipse Maven project archive | 32 lines

RandomDataGenerator.java

FILE 30 OF 32

src/test/java/utilities/RandomDataGenerator.java

```
package utilities;

import java.util.Random;

public class RandomDataGenerator {

    public static String getUsername() {

        return "user" +
            System.currentTimeMillis();
    }

    public static String getPassword() {

        return "Test123";
    }

    public static String getPhone() {

        Random random = new Random();

        return "98" +
            (10000000 + random.nextInt(9999999));
    }
}
```

Source: Eclipse Maven project archive | 25 lines

ScreenshotUtil.java

FILE 31 OF 32

src/test/java/utilities/ScreenshotUtil.java

```
package utilities;

import java.io.File;
import java.io.IOException;
import java.nio.file.Files;
import java.nio.file.StandardCopyOption;

import org.openqa.selenium.OutputType;
import org.openqa.selenium.TakesScreenshot;
import org.openqa.selenium.WebDriver;

public class ScreenshotUtil {

    public static String captureScreenshot(WebDriver driver,
                                           String testName) {

        String path = System.getProperty("user.dir")
            + "/reports/screenshots/"
            + testName + ".png";

        try {

            File src = ((TakesScreenshot) driver)
                .getScreenshotAs(OutputType.FILE);

            File dest = new File(path);

            dest.getParentFile().mkdirs();

            Files.copy(
                src.toPath(),
                dest.toPath(),
                StandardCopyOption.REPLACE_EXISTING);

        } catch (IOException e) {

            e.printStackTrace();

        }

        return path;

    }
}
```

Source: Eclipse Maven project archive | 42 lines

WaitUtils.java

FILE 32 OF 32

src/test/java/utilities/WaitUtils.java

```
package utilities;

import java.time.Duration;

import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.support.ui.WebDriverWait;
import org.openqa.selenium.support.ui.ExpectedConditions;

public class WaitUtils {

    public static void waitForElement(WebDriver driver,
                                      WebElement element) {

        WebDriverWait wait =
            new WebDriverWait(driver,
                             Duration.ofSeconds(10));

        wait.until(ExpectedConditions.visibilityOf(element));
    }
}
```

Source: Eclipse Maven project archive | 21 lines